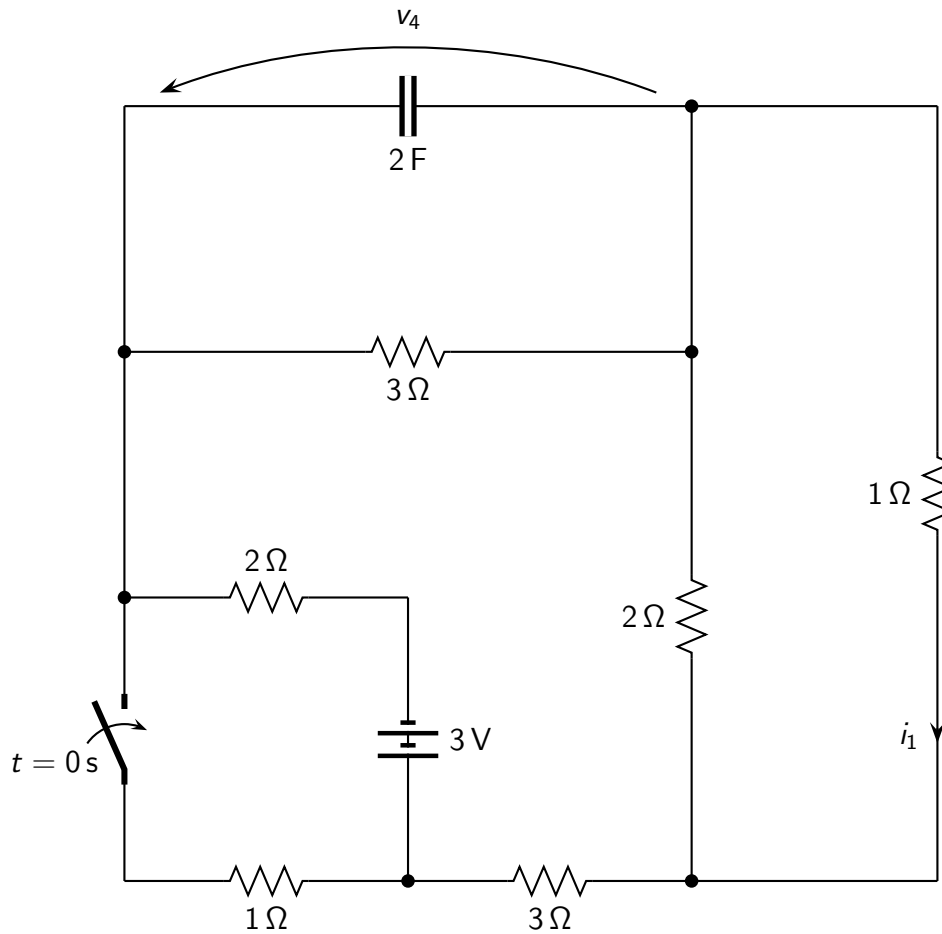


Problem: find the expression of $i_1(t)$, $v_4(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{39}{11} \text{ s}$;
- Initial conditions before switching: $v_4(0^-) = \frac{-27}{26} \text{ V}$;
- Initial conditions after switching: $i_1(0^+) = 5.917 \text{ mA}$; $v_4(0^+) = \frac{-27}{26} \text{ V}$;
- Steady-state solution: $i_1(\infty) = \frac{-1}{11} \text{ A}$; $v_4(\infty) = \frac{-9}{22} \text{ V}$;
- Solution for $t > 0$:

$$i_1(t) = \left(96.826 \cdot 10^{-3} e^{-t/\tau} + \frac{-1}{11} \right) \text{ A}$$

$$v_4(t) = \left(-0.629 e^{-t/\tau} + \frac{-9}{22} \right) \text{ V}$$